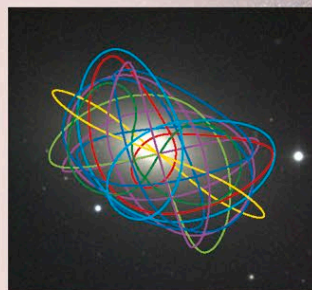


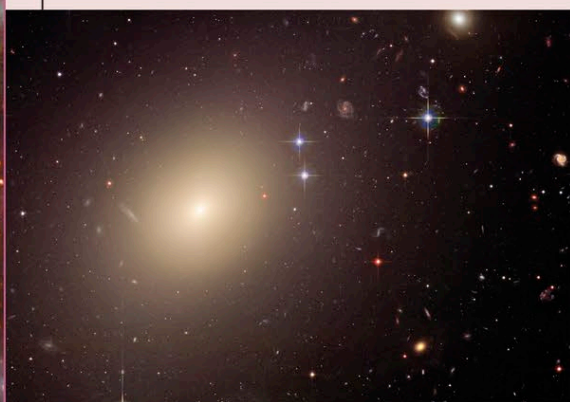
ELLIPTICAL GALAXIES

Elliptical galaxies show little structure other than a simple ball shape. They span the range from the largest to the smallest galaxies. At one end, dwarf ellipticals are relatively tiny clusters of a few million stars, often very loosely distributed, appearing faint and diffuse. Such galaxies are scattered in the space between larger galaxies and must contain significant amounts of invisible material simply to hold them together. Some of this could be in a central black hole, but much of it seems to be mysterious “dark matter” (see p.27) scattered through the whole of the galaxy. At the other extreme lie the giant ellipticals—galaxies only found near the centers of large galaxy clusters and often containing many hundreds of billions of stars. Some giant ellipticals, called cD galaxies, have large outer envelopes of stars and even multiple concentrations of stars at their centers, suggesting they may have formed from the merging of smaller ellipticals. Almost all the stars in elliptical galaxies are yellow and red, and there is rarely any sign of star-forming gas and dust. The dominance of old, long-lived stars implies that any star formation in these galaxies has long since ended. Each star orbits the galaxy’s dense core in its own path. The chances of collision are very remote, because stars are so small relative to the distances between them. With no gas and dust clouds to interact with, there is nothing else to flatten the stars into a single plane of rotation. Ellipticals are described according to their degree of elongation—how much they deviate from a perfect sphere (see p.302)—but the largest galaxies are always very close to perfect spheres.



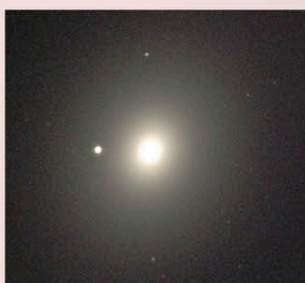
ORBITS IN ELLIPTICAL GALAXIES

The orbits of stars in an elliptical galaxy vary wildly, from circles to very long ellipses, and are not confined to any specific direction.



GIANT ELLIPTICAL

ESO 325-G004 is a giant elliptical galaxy at the heart of Abell S740, a cluster 450 million light-years away in Centaurus. Thousands of globular star clusters form pinpoints of light around the central galaxy.



INTERMEDIATE GALAXY

M49 in the Virgo Galaxy Cluster is a large elliptical of type E4. With a diameter of about 160,000 light-years, it is classed by some astronomers as a giant elliptical, although its mass is much less than that of the true giants.



DWARF ELLIPTICAL

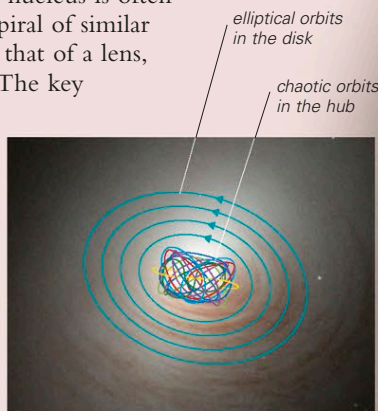
The Leo I Galaxy is a nearby dwarf elliptical and one of the few we can study closely. With so few stars, there must be a large amount of dark matter holding the galaxy together with its gravity.

LENTICULAR GALAXIES

At first glance, lenticular galaxies appear to be relatives of ellipticals—they are dominated by a roughly spherical nucleus of old red and yellow stars. However, around this nucleus, these galaxies also have a disk of stars and gas. This links them to spiral galaxies, and they are similar in overall size and general shape, although the nucleus is often considerably bigger than it would be in a spiral of similar size. The overall shape is often described as that of a lens, which is the root of the name “lenticular.” The key difference between lenticulars and spirals is that lenticulars have no spiral arms and little sign of star-forming activity in their disks. Without the bright blue star clusters that illuminate the disks of spirals, lenticulars are sometimes hard to tell apart from ellipticals. Those that are face-on may be indistinguishable from ellipticals and misclassified. An edge-on spiral galaxy with a large nucleus can equally be misclassified as lenticular, because at oblique angles spiral structure is often invisible. Astronomers are uncertain how lenticular galaxies form, but they could be spiral galaxies that have lost most of their dust and gas.

DUSTY LENTICULAR

Lying 25 million light-years away, galaxy NGC 2787 is one of the closest lenticular galaxies. Dust lanes can be seen silhouetted against the nucleus, marking the plane of its disk.



ORBITS IN LENTICULAR GALAXIES

Stars in the nucleus of a lenticular galaxy follow orbits with no specific plane, similar to those in an elliptical galaxy or a spiral nucleus. Gas and dust in the disk orbits in a more orderly plane.

